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Review

Delisting the Northern Rocky Mountain gray wolf from the US Endangered Species Act: an assessment of political discourse over 20 years

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Feared, revered, and politicized, wolves have long captured human imagination, and ignited fierce conservation conflicts. In the United States, the Endangered Species Act protects species at risk of extinction from human impacts. This far-reaching legislation, which impacts development and state-level wildlife management, has been fraught with legal battles and controversy. The gray wolf, first listed in 1974, exemplifies the tumultuous back-and-forth of listed species. In 2011, Congress, rather than wildlife management agencies in the executive branch of government, delisted the gray wolf in two Northern Rocky Mountain states using legislative action, specifically a rider on a budget bill. Since then, gray wolf territory has expanded, and controversy over gray wolf management prevails. Our study used quantitative structural topic modeling to assess political discourse surrounding the Rocky Mountain gray wolf delisting in newspapers and other documents between 2005 and 2025, allowing us to assess how different stakeholders discuss the gray wolf and how language has shifted over time. In our study, we find that nongovernmental organizations are generally the most vocal in gray wolf discourse. Further, our results show that the 2011 delisting shifted the language used by a variety of stakeholders when debating gray wolf management, with the 2011 delisting being used as both a cautionary example and a model to follow.

Keywords: carnivore management policy, Endangered Species Act, political discourse, wolf delisting

Introduction

Few pieces of environmental legislation in the United States (US) have provoked as much sustained debate as the Endangered Species Act (ESA) of 1973 (16 USC §1531 et seq.). Enacted in response to accelerating biodiversity loss driven by development



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and economic growth, the ESA is a powerful statutory mechanism for conservation that protects species at high risk of extinction and the habitats essential to their survival (Czech and Krausman 2001, Bean 2009). The ESA establishes a science-based process for identifying species at risk, listing them as endangered (e.g. at risk of extinction) or threatened (e.g. likely to become endangered in the near future), and ultimately guiding their recovery and delisting (Taylor et al. 2005, US Fish and Wildlife Service n.d.). Federal agencies, the US Fish and Wildlife Service (USFWS) and the National Marine Fisheries Service, are required to develop and implement recovery plans with the ultimate goal of delisting species once populations are deemed viable (Cubbage et al. 2016, Gerber et al. 2018, Haines et al. 2021). Unlike many environmental statutes, the ESA applies to public and private lands, regulating federal actions as well as private development (Cubbage et al. 2016), and this authority has made the ESA a point of political and legal controversy. This tension is exemplified by the logging restrictions imposed to protect the northern spotted owl in the Pacific Northwest (Noon and Blakesley 2006) and the legal battle over the snail darter and the Tellico Dam project in Tennessee (Perry 2012). The ESA also sparks conflict when the protected status of a species generates disagreement among stakeholder groups. For example, ESA-listed carnivores such as grizzly bears *Ursus arctos horribilis* and gray wolves *Canis lupus* often evoke deeply held values among different stakeholder groups, but they also prey on wildlife and livestock, leading to persistent conflicts with hunters and livestock producers (Fitzgerald 2011).

The relationships between wolves and the ecosystems they inhabit are complex (Smith and Peterson 2021), as are the public's and policymakers' values and views regarding how best to co-exist with these animals (Nie 2003). Due to this complexity, wolves have been the source of tumultuous controversy between branches of government, opposing stakeholder groups, and the federal and state governments in the US (Martin 2020). By the mid-20th century, gray wolves had been nearly eradicated from the US due to government-led culling and habitat loss, leading to gray wolves being listed under the ESA (1974). In 1995, as part of their ESA recovery plan, gray wolves were reintroduced to Yellowstone National Park and central Idaho with the goal of establishing 10 breeding pairs (Guertin 2016).

Gray wolves in the Northern Rocky Mountain states of Wyoming, Idaho and Montana have an especially interesting and politically charged history under the ESA. In 2009, the USFWS determined the population of gray wolves within the Northern Rocky Mountain region qualified to be managed under the ESA as a distinct population segment (DPS) because of their distinctive geographic distribution, population ecology, and genetics (Fig. 1) (Endangered and Threatened Wildlife and Plants). Under the ESA a DPS is defined as vertebrate population discrete from other populations of that species (NOAA Fisheries 2022).

After being distinguished as a DPS, gray wolves in this region were considered recovered, based on conditions outlined in a 1987 Northern Rocky Mountain Recovery Plan (US Fish and Wildlife Service 1987). Subsequently, the

USFWS proposed delisting of this population of gray wolves in 2009, a decision that was reversed in 2010 (Robbins 2009, US Fish and Wildlife Service 2011). During this period, wolf delisting became a prominent issue in federal legislative campaigns. In Montana, Senator Jon Tester (a Democrat) emphasized his support for delisting as part of his reelection campaign against Republican Denny Rehberg, a vocal critic of wolf protections under the ESA (Peoples 2011). In 2011, a budget bill was passed by Congress that included a controversial rider legislatively delisting the gray wolf in much of the Northern Rocky Mountains. A rider is a provision added to a larger bill that is likely to pass, such as a budget bill. Often, riders are unrelated to the main subject and used to pass politically divisive legislation. This provision allowed for gray wolves in Montana and Idaho to be delisted, and later Wyoming, while nationwide wolves remained on the endangered species list. It is worth noting that in 2024, the USFWS determined that the Northern Rocky Mountain population of gray wolves no longer constituted a DPS (89 FR 8391 2024). Although they are no longer considered a DPS, in 2025, gray wolves remain delisted from the ESA in the Northern Rocky Mountain states.

The 2011 congressional delisting of gray wolves was notable because it bypassed the typical ESA process. Rather than the USFWS leading the delisting through scientific and administrative review, Congress removed wolves from the endangered species list by legislative action. This outcome marked a significant shift in wolf politics, elevating congressional actors as key decision-makers in the future of wolf recovery. Researchers have long debated the role of politics in the ESA (Carolan 2008), arguing that this event was not the first time that politics influenced the type of science and outcomes of ESA decisions (Wyman 2012). Nevertheless, delisting the Northern Rocky Mountain gray wolf illustrates an increasing phenomenon of politicians overtly wielding power over an endangered species' fate, rather than scientists (Mollett et al. 2025). Given these political shifts, our research aims to better understand the political discourse surrounding gray wolf delisting, using the 2011 Northern Rocky Mountain population decision as a key shifting point. To achieve our research goals, we assess which stakeholder groups – such as federal and state wildlife agencies, legislators, lawyers/judges, NGOs, and members of the public – were most vocal in shaping the conversation surrounding delisting. We also examine how rhetoric differed between stakeholder groups, and how patterns evolved across distinct phases of the delisting timeline.

We systematically reviewed the political discourse in nationwide newspaper articles and media coverage of the Northern Rocky Mountain gray wolf delisting process between 1 January 2005 and 1 March 2025. While we focused on articles that discuss the Northern Rocky Mountain gray wolves, we included all national news sources because this topic has drawn broad attention. This method allows us to assess the dialogue and evaluate streams of political discourse regarding this illustrative example of political involvement in ESA decision-making. We focused on the Northern Rocky Mountain states of Montana, Idaho and Wyoming from the five years before delisting to early 2025. Our research



Figure 1. Map of the Northern Rocky Mountain gray wolf DPS and current (2025) gray wolf presence in the northwestern US.

is timely because gray wolf management continues to be a politically divisive and frequently debated topic in the intermountain West region of the US. For example, in Idaho, gray wolf populations currently exceed target populations set by state agency managers in wildlife plans. Some interest groups, such as livestock producers, are concerned about livestock depredation from these higher than-anticipated wolf populations (Congressional Sportsmen's Foundation 2024, Corbin and Druzin 2025). Additionally, Colorado reintroduced gray wolves as an experimental population using a ballot referendum. Wolf reintroduction in Colorado has led to conflicts between user groups and debates similar to those surrounding delisting of the Northern Rocky Mountain subpopulation in 2011 (Niemiec et al. 2020, Gonzalez et al. 2024, Robbins 2024, Gabehart 2025).

Material and methods

In this study, we employed a three-step method using an adaptive multi-stakeholder governance theoretical lens,

looking closely at stakeholders who are vocal in ESA decision-making as our scope. First, we conducted a document review to collect all media coverage of the Northern Rocky Mountain gray wolf population between January 2005 and March 2025. Next, we reviewed all relevant articles using structural topic modeling (STM) to identify words that were co-occurring between political actors and across time. Finally, the STM output was compared qualitatively to the document review spreadsheet to apply narrative descriptions to the co-occurring word-sets and identify nuances in the quotes used by political actors across time, focusing on differences in political discourse before, during and after listing.

Document review: defining stakeholders in our adaptive governance lens

Adaptive governance highlights the importance of incorporating science and other knowledge types to effectively manage and create policy in complex environmental and natural resource systems (Brunner 2005, Chaffin et al. 2014). Previous studies have applied adaptive governance principles

to complex ESA conflicts, using adaptive multi-stakeholder governance to describe the political discourse surrounding grizzly bear delisting in the Greater Yellowstone Ecosystem (Mollett et al. 2025). The adaptive governance framework begins with the idea of environmental governance, defined as political actors (e.g. agencies and stakeholders) using scientific information to implement policy that impact ecosystems and species. Three conditions lead to environmental governance becoming adaptive governance, including: 1) dialogue between stakeholders, 2) inclusion of all relevant institutions, ranging from the private sector, community groups, to government at all scales, and 3) experimentation with the way that policymakers make and implement actual policy (Morris et al. 2024).

In this study, we adopted a similar methodology to a previous study assessing the co-occurring words surrounding grizzly bear delisting (Mollett et al. 2025). We call these co-occurring word-sets 'threads of political discourse'. Our unit of analysis is decision-making stakeholders (e.g. political actors), defined broadly as government employees and other active voices in decision-making surrounding gray wolf delisting in three Northern Rocky Mountain states (e.g. Montana, Idaho and Wyoming). For the purpose of this study, we categorize political actors into five groups of stakeholders who are publicly speaking out and encouraging changes to policy: executive branch, legislative branch, judicial members, members of the public or private sector representatives, and non-governmental organizations (NGOs). The executive branch includes federal land management agencies (e.g. USFWS, or the general Department of the Interior), state wildlife management agencies (e.g. Montana Department of Fish, Wildlife, and Parks; Idaho Fish and Game; Wyoming Game and Fish), state governors and their spokespeople, and tribal wildlife management departments, or councilmembers. The legislative branch category includes members of state legislatures as well as members of Congress. The judicial members refer to attorneys general in Montana, Wyoming, Idaho and US district judges. Members of the public and private sector representatives include members of the public, retired scientists, authors, and researchers at universities. Finally, the NGO category generally refers to NGO directors, legal advocates, or attorneys affiliated with NGOs.

Document review: article selection process

To assess the threads of political discourse surrounding gray wolf delisting in the Northern Rocky Mountain states, we used the US Newsstream Proquest Database to systematically review media coverage (e.g. newspapers, magazines, press releases) published between 1 January 2005 and 1 March 2025. In this search, we used 'wolf delisting' with boolean operators to specify the state or region (i.e. 'Rocky Mountain', 'Idaho', 'Montana' and 'Wyoming'). We determined the term 'delisting' to be appropriate after exploring broader 'gray wolf' and 'Endangered Species Act' searches which yielded thousands of irrelevant articles. These four search terms yielded 786 articles, 207 of which were relevant and reviewed in further detail. We considered an article relevant if it was

about gray wolf delisting and discussed the Rocky Mountain subpopulation beyond just naming the topic (i.e. articles that said Rocky Mountain population but were about the Great Lakes were not included). In the data cleaning process, we removed an additional 10 articles because they were duplicate articles printed in different news sources with unique titles.

In the relevant articles, quotes by political actors regarding wolves were extracted and recorded, along with the name of the political actor. Prior to analyses, we split the articles into three periods: before (e.g. 2005–2010), during (e.g. 2011), and after (e.g. 2012–2025) the monumental Congress-initiated delisting decision.

Data analysis

We used structural topic modeling (STM) to analyze political discourse surrounding gray wolf delisting, focusing on how patterns of language varied across two key dimensions: political actor identity and time period. Specifically, we modeled quotes extracted from media coverage and attributed to political actors, allowing us to compare the prevalence of discourse threads across five actor categories (executive, legislative, judicial, NGO and public/private sector representatives). We also analyzed the entire text of the article in three time periods (pre-delisting, during delisting and post-delisting). Our goal was to assess how the content and emphasis of gray wolf discourse evolved across stakeholder groups and over time. We first pre-processed the text using a custom stopword list to remove frequently occurring but uninformative words (get, like, one, re, said/say/says, will, wolf/wolves, gray, Wyoming, Idaho, Montana), implemented with the 'SnowballC' R package (Bouchet-Valat 2023). To ensure model quality and interpretability, we retained only terms that appeared in at least 10 articles (Roberts et al. 2019). We ran our STM using spectral initialization for 20 replicates, with a maximum of 10 000 iterations to ensure convergence and model robustness (Hosseiny Marani and Baumer 2023). Additionally, we evaluated topic quality using semantic coherence and exclusivity. Semantic coherence reflects the extent to which top words in a topic co-occur in articles, whereas exclusivity indicates how uniquely those words are associated with a single topic (Airoldi and Bischof 2016, Lee and Mimno 2017, Weston et al. 2023).

We identified ten co-occurring word sets, five threads for political actors, and five for time-periods, and conducted means parameterized regressions to assess the prevalence of each thread by group (i.e. political actors and time-periods). Using a means parameterized regression allowed us to directly estimate the average effects of groups on the prevalence of each thread of political discourse. We used the *estimateEffect* function in the 'STM' package to fit two models with thread of discourse prevalence as the dependent variable, and either 'political actor' (i.e. executive, legislative, judicial, NGO, or members of the public and private sector representatives) or 'time-period' (i.e. before, during, after) as the independent variable (Roberts et al. 2019). To identify significant differences in the frequency of threads for each group, we compared the mean 95% confidence interval around the coefficient estimates.

Table 1. Distribution of quotes by political actor categories.

Political actor category	Example roles	No. of quotes in category	Percent of total
NGO	• General NGO representatives • Directors • Attorneys or legal advocates	292	52%
Executive branch	• Federal agencies • State agencies • State governor, or representative • County commissioners • Tribal chairpeople and wildlife program directors	144	26%
Legislative branch	• State legislature • Congress	88	15%
Members of the public and private sectors	• The general public • University researchers	28	5%
Judicial branch	• US District Judge • State Attorney General	14	2%
Total		566	100%

While STM allows for unique evaluations of political discourse across many articles, the results can sometimes be complex to interpret. Following the STM analysis, we qualitatively reviewed the quotes and full articles to assess the co-occurrence outputs, create narrative descriptions, and validate the STM results (Lindstedt 2019, Gillings and Hardie 2023). Our follow-up qualitative assessment also helped to add context to the themes presented in the co-occurrence analysis.

Results

Quantitative analysis overview

We closely reviewed 197 articles, using the full text to assess discourse over time, and selected all statements made by political actors. The analysis of quotes by political actors yielded 566 relevant quotes, 467 unique words and 3734 total words. The time-period full article text assessment included 197 full-text articles, 976 unique words, and 7785 words. Commonly occurring, but uninformative words (e.g. get, like, will) were removed prior to analysis.

The distribution of our 566 individual quotes varied across the five categories of political actors (Table 1). Quotes made by the same political actor in separate articles were considered unique quotes in Table 1 whereas multiple quotes from one actor in the same article are only counted once. The results of our review showed that NGOs or political advocates (i.e. attorneys for NGOs) were the most vocal in the media coverage surrounding the delisting of the Rocky Mountain gray wolf subpopulation (52%; $n=292$), followed by the executive branch (26%, $n=144$). The legislative branch also comprised 15% ($n=88$) of the 566 identified quotes coming from this group. Less frequently quoted were members of the public and private sector representatives (5%, $n=28$) and the judicial members (2%, $n=14$).

The reviewed articles were relatively evenly distributed between periods, despite an uneven distribution of years (Fig. 2). In the six years leading up to the 2011 delisting decision, 68 articles (34%) were published discussing delisting

the Northern Rocky Mountain gray wolf population. In 2011 alone, 49 articles (25%) were published. In the 14 years after the delisting, 80 articles (41%) were published discussing wolf delisting and the Northern Rocky Mountain gray wolf population.

Political actors most vocal in decision-making: distribution of co-occurring words

Our analysis produced five threads of political discourse across political actors: thread 1) state management plans as a viable way to manage wolves (rather than federal management under the ESA); thread 2) state management by way of hunting seasons and lethal removal for lessening human wildlife conflict; thread 3) status of ESA protections, focusing on technical debates between stakeholders over whether they see wolves as being scientifically recovered; thread 4) continued questions over whether federal or state agencies should be responsible for wolves; and thread 5) questions over politicization of decisions on the ESA. These five topics exhibited strong interpretability and topic separation. Semantic coherence scores ranged from -87.50 to -55.16 (Table 2), where values closer to zero suggest the most probable words within each topic tended to co-occur in the same articles, increasing interpretability. Exclusivity scores ranged from 8.28 to 9.34 (Table 2), indicating that high-probability words were relatively unique to individual topics and not shared across multiple themes.

Our analysis shows that the different political actor groups expressed varying concerns and positions when quoted in news sources (see Fig. 3 and Table 2 for threads summarized with example quotes). We anticipated the executive branch, with management authority for listed species, to be the major political voice across all five threads of discourse. However, we found that the executive branch was only identified as the most frequent voice when describing the utility of state management plans (thread 1; estimated mean prevalence (EMP) = 0.410, $p < 0.001$). Specifically, articles quoted state governors' offices and state wildlife management agencies discussing their state management plans, or those of the other

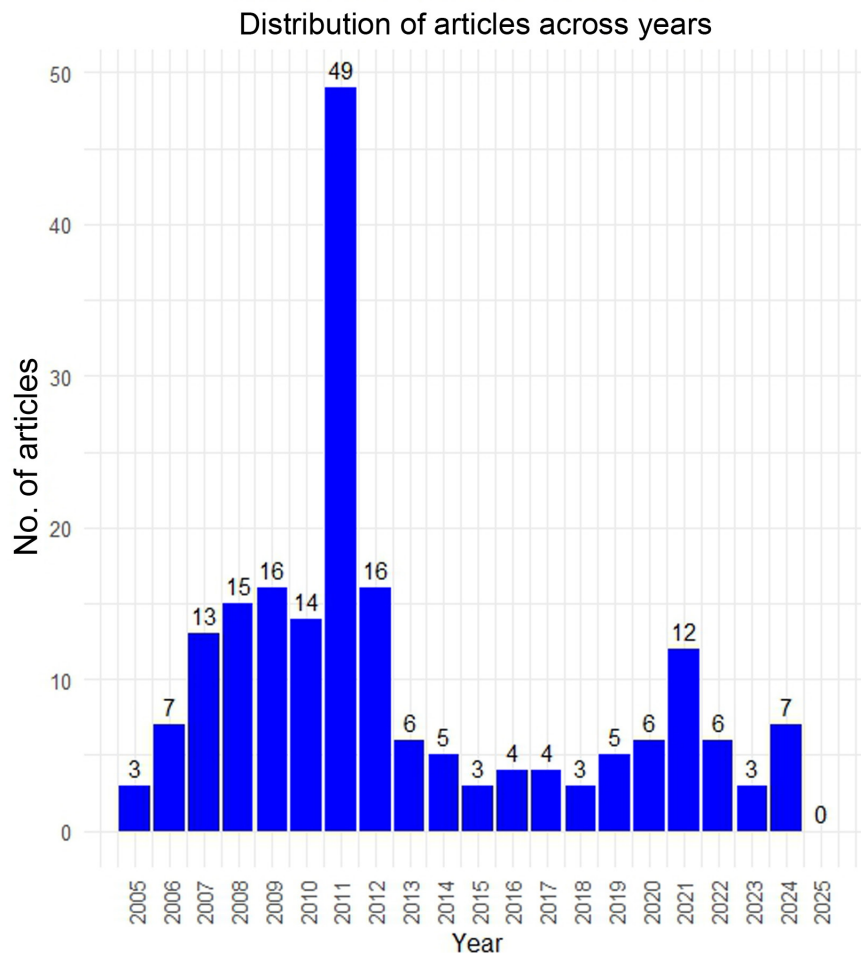


Figure 2. Distribution of articles across time 2005–2025. Note: Document collection ended in March 2025. At this time there were no articles published.

northern Rocky Mountain states. In addition to the executive branch, the legislative branch was vocal in this thread ($EMP=0.379$, $p < 0.001$), discussing the significance of state-led plans for managing animals within state boundaries (Table 2). Finally, members of the public and private sector representatives were also frequently quoted ($EMP=0.330$, $p < 0.001$) expressing interest in state management of wolves to mitigate impacts on wildlife and livestock. Political actors in this thread advocated for state management of wolf populations to reduce conflicts with other wildlife, highlighting state plans for managing wolf populations. They focused on states' capacities, authority to manage wildlife within their jurisdiction, and the ability of states to manage wolf populations sustainably.

Our second thread of political discourse (thread 2) focused on state management to allow for a regulated hunting season as well as allowing lethal management for ranchers and others whose livelihoods may be affected by wolf depredation. Rather than describing management plans, this thread focused more on the use of hunting seasons or lethal management as a strategy to prevent human–wildlife conflict, where livestock is killed by wolves. While most political actors engaged in

this thread, the most vocal political actors were the members of the public and private sectors ($EMP=0.260$, $p < 0.001$). Often, political actors in this thread were ranchers or hunters at public meetings about wolf management. These stakeholders expressed frustration with conflicts between wolves and livestock and competition between wolves and hunters for ungulate populations and often appealed to delisting as a tool to minimize losses. Judicial actors also frequently engaged in this thread ($EMP=0.234$, $p < 0.001$). To a lesser extent the executive branch ($EMP=0.208$, $p < 0.001$) and NGOs ($EMP=0.166$, $p < 0.001$) also engaged in discourse surrounding hunting and lethal management.

Thread 3 of political discourse captured the debate over the status of ESA protections. This thread was concerned with the scientific information available, questions of connectivity between populations, and how many animals or breeding pairs constitute recovery. This group's most frequent political actors were the judicial members ($EMP=0.519$, $p < 0.001$), often quoted discussing USFWS decisions during one of the multiple delisting and relisting decisions made between 2005 and 2025. The legislative branch was also vocal in this thread ($EMP=0.435$, $p < 0.001$); representatives from this group

Table 2. Sets of co-occurring words, most prominent voices, and narrative descriptions for political actors. ¹The political actors exemplified in this column represent the top political actors with a significant number of quotes in that thread.

Thread of political discourse	Semantic coherence (exclusivity)	Narrative description	Most common political actor(s) ¹	Example quote(s)
Thread 1: manage, state, population, wildlife, plan, work, delist	–62.83221 (8.284126)	State management — plan in place: This thread highlighted the utility of state plans and argued that they can work and should be the primary approach due to wolf impacts on wildlife. It also contained discussions of the content of state plans.	Executive, legislative, members of the public and private sectors	‘The governor is confident it will meet its obligation under the management plan to sustain the required number of wolves. Our plan will maintain a sustainable population of wolves in Northwestern Wyoming, and it will contribute Wyoming’s share of the recovery goal of wolves in the Rocky Mountains.’ – Executive branch; Steve Ferrell, wildlife policy adviser to Governor Mead (R-WY) (Feds Prepare to End Wolf Protections in Wyoming 2012) ‘Wolves have recovered in the Northern Rockies. By untying the hands of the Montana biologists who know how to keep the proper balance, we will restore healthy wildlife populations, and we will protect livestock. This provision is best for our wildlife, our livestock, and for wolves themselves.’ – Legislative branch; Senator Jon Tester (D-MT) (Federal Budget Deal Strips Protection from Water, Wild Lands, Wolves 2011) ¹ Idahoans know that the state is more than prepared to manage our wolf population’ – Legislative branch; Senator Mike Simpson (R-ID) (Wolf Issue Shows Need for ESA Reform’ 2011). ‘It should be the people in Idaho deciding whether we have wolves or not’ – Member of the public/private sector; Rex Rammell, a veterinarian, former elk rancher (Knickerbocker 2008) ‘I came because I want to protect our wildlife. I don’t care what the pro-wolf people say, these wolves were never here before they reintroduced them in 1995... Vail said during his years tending to a nearby ranch, he has seen his fair share of wolf-kill gore... Vail said he plans on getting his wolf hunting tags next year. He had a tag in 2009 but was unable to fill it before the short season ended.’ – Member of the public; Ray Vail Gallatin Gateway Rancher (Keller 2011)
Thread 2: population, state, hunt, kill, managing, wildlife, area	–75.50905 (8.780706)	State management – Hunting season and lethal control: This thread focused on wolves’ impacts on other species and the values attributed to wolves. Actors often advocated hunting seasons and lethal control for livestock producers to mitigate conflicts between wolves and livestock or wildlife.	Members of the public and private sectors, judicial, executive, NGOs	‘Called the wolves delisting arbitrary and capricious and said it demonstrated a possibility of irreparable harm to the species. The wildlife service provides no new evidence or research to support its change of course, Molloy wrote. Congress does not intend agency decision-making to be fickle... Genetic exchange has not taken place and is in fact rare, the judge wrote. He cited a 2007 study commissioned by the wildlife service itself. Genetic exchange that has not taken place between larger subpopulations under [ESA] protections is not likely to occur with fewer wolves under state management.’ – Judicial member; US District Judge Donald W. Molloy of Missoula (Abdollah 2008). ‘No one can rationally argue that the Rocky Mountain gray wolf is still endangered. Wolf populations in the west are robust and far exceed recovery goals... If the [ESA] is going to be effective at all, we need to remove recovered species from the list and consider it a victory.’ – Legislative branch; Representative Mike Simpson (R-ID) (US Newsstream Collection 2011)
Thread 3: species, endangered, act, delist, secretaries, success, interior	–66.27771 (9.344163)	Status of ESA protections: This thread discusses the technical components of gray wolves’ status under the ESA, the ESA’s role in management, and whether the wolf populations have reached delisting levels.	Judicial, legislative	

(continued)

Table 2. Continued.

Thread of political discourse	Semantic coherence (exclusivity)	Narrative description	Most common political actor(s) ¹	Example quote(s)
Thread 4: wildlife, protect, service, recovery, fish, federal, state	–55.16080 (8.972741)	Questioning recovery management responsibility: This thread focused on questioning the recovery of the gray wolf and debating whether the USFWS should have management responsibility or if the states can be trusted to manage wolves.	NGO	‘Ultimately, ensuring the protection of vulnerable species like wolves requires comprehensive conservation strategies, including stronger regulation enforcement and reconsideration of management responsibilities. Relisting wolves under the [ESA] is necessary to address the shortcomings of state management and prevent further harm to these iconic animals... Wyoming, Montana, and Idaho have shown they cannot be trusted when it comes to sustaining wolves and other rare and sensitive species.’ – NGO; Jessica Karjala, Footloose Montana (Karjala 2024) ‘We all need to work together to craft responsible state management plans for wolves that allow for healthy interconnected wolf populations now and in the future.’ – NGO; Rodger Schlickseisen, President of Defenders of Wildlife (Federal Protections Restored for Northern Rockies’ Wolves 2010) ‘We will not allow the fate of endangered species to be determined by politicians serving special interests’ – NGO; Michael Garrity, Alliance for the Wild Rockies (Environmental Groups Sue Government over Wolf Delisting 2011). ‘But while former USFWS director Dan Ashe agreed wolves have sufficiently recovered for delisting, he questioned the announcement coming so close to the election. It creates the perception that it’s being done for political reasons.’ – Public/private sector; Dan Ashe, former USFWS director (Nichols 2020)
Thread 5: director, group, executive, delist, nature, political, plan	–87.50037 (8.929439)	Questioning decision makers: Actors expressed uncertainty about the USFWS’s decisions and politics’ role in government decision-makers’ delisting the gray wolf.	NGO, members of the public and private sectors	

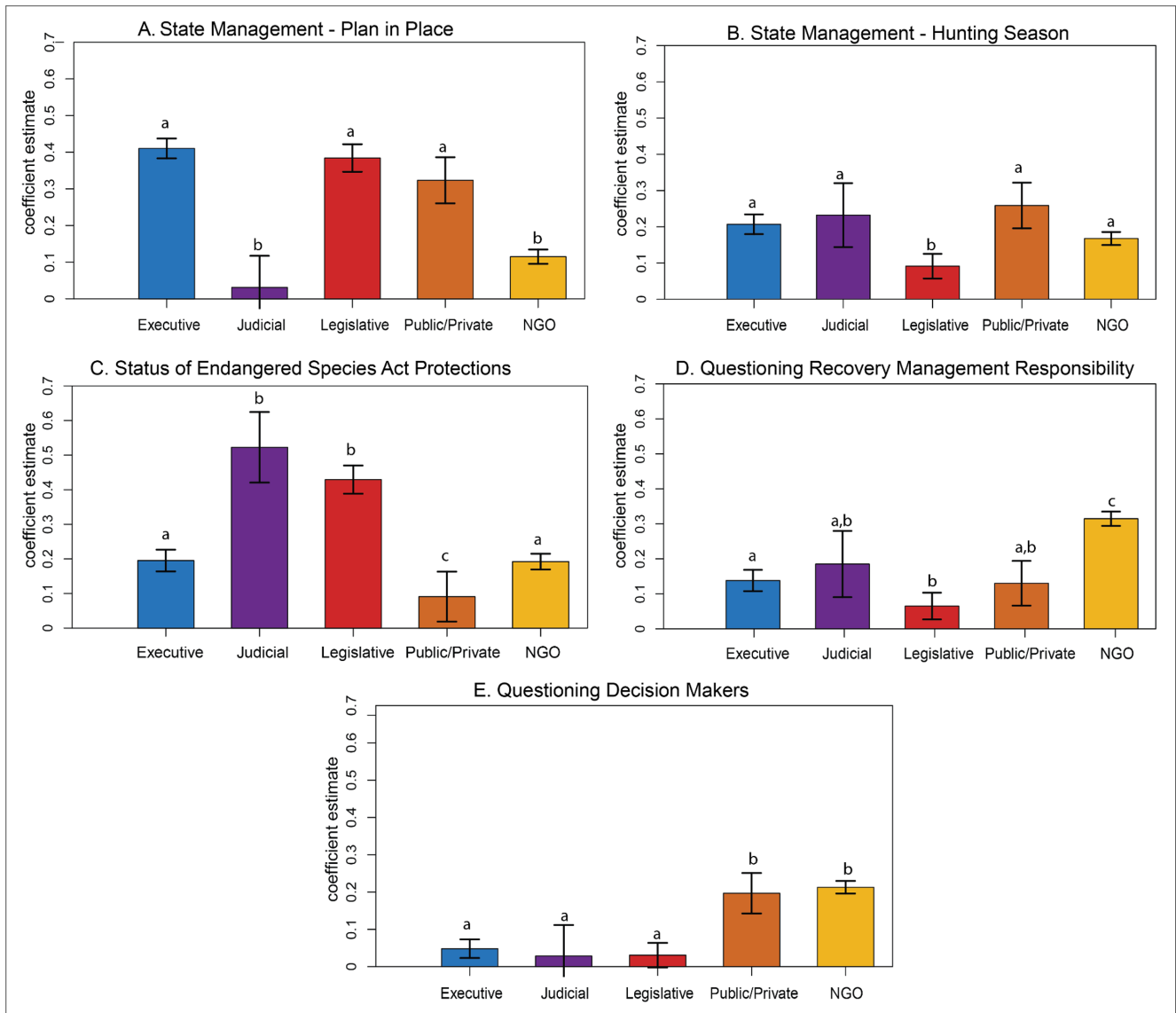


Figure 3. Predicted frequency of use of the five most common threads by political actors derived from the means parameterization regression. The bars represent the predicted mean frequency of the use of each topic (i.e. the coefficient estimate from the regression) for each political actor. Confidence intervals around each bar indicate the 95% confidence interval around these mean frequencies. Letters above the bars denote statistically significant differences among political actors. Each panel (A–E) denotes the frequency of the statistical topic where the model identified co-occurring words for the dataset as a whole, and panels are labeled by the qualitative interpretation of these themes.

highlighted delisting as an indicator of the success of the ESA in reviving wolf populations (Table 2).

Thread 4 involved questions regarding who should be responsible for managing wolves: states or the federal government. NGOs (EMP=0.311, $p < 0.001$) were most likely to debate whether the gray wolf population had reached a point where delisting and turning management over to the states was prudent. Some groups expressed frustration with the USFWS for putting management of wolves into the hands of the Northern Rocky Mountain states, due to concerns over the goals of these states (Table 2).

Thread 5 mainly consisted of NGOs (EMP=0.214, $p < 0.001$) and members of the public or private sector

representatives (EMP=0.188, $p < 0.001$) questioning decision-makers' motivations in these wolf delisting decisions. Quotes within this thread expressed concern that decisions may have been politically rather than scientifically motivated, voicing skepticism regarding the timing of wolf delisting decisions so close to elections (Table 2).

Distribution of co-occurring words before, during, and after 2011 delisting

We identified five unique threads of temporal political discourse across time through a review of full article texts. In this section 197 articles were analyzed and categorized into three groups: before ($n = 69$, 34%); during ($n = 49$; 25%); and after

delisting ($n = 80, 41\%$). Thread 1 of the temporal model contained state management and the content of plans. Thread 2 described discourse surrounding state management by way of hunting seasons and lethal removal for lessening human–wildlife conflict. In thread 3, articles discussed delisting as an ESA success story. Thread 4 described the debate surrounding how delisting was carried out, and thread 5 involved management responsibility and state protections. The five topics demonstrated strong interpretability and topic separation, with semantic coherence values that ranged from -26.70 to -7.53 , where higher values (closer to zero) indicate that the top words in a topic tend to co-occur in articles, supporting interpretability. Exclusivity values ranged from 7.55 to 8.58 , suggesting moderate to high uniqueness of terms across topics. All temporal political discourse threads are described with example quotes in Table 3 and visualized in Fig. 4.

Some threads were more of a focus during specific time periods. For example, our assessment showed that before the 2011 delisting decision, the temporal political discourse focused on the adequacy of state management plans to take over wolf management following a possible delisting (thread 1). During this time-period, political actors focused on the content of state management plans, discussing whether they provide sufficient protection ($EMP = 0.585, p < 0.001$). Specifically, before delisting, the content of the state of Wyoming's plan was a source of concern for NGOs and executive branch actors.

State management of wolves using hunting and lethal control (thread 2) did not vary significantly before, during, or after the 2011 delisting. The concern here surrounds the livelihoods of hunting outfitters and guides, and livestock producers. Both stakeholder groups expressed concerns about wolves killing ungulates and livestock, and their lack of recourse through hunting or lethal control of predating wolves. Other stakeholders lamented before the 2011 decision that delisting and turning wolves over to state management was overdue.

Surrounding the Congressional ESA delisting in 2011, the temporal political discourse concerned the success of the ESA in allowing agencies to reach the point of delisting (thread 3) ($EMP = 0.550, p < 0.001$). During this time-period political actors frequently described this decision as a success of scientists, wildlife managers, and stakeholders in recovering gray wolves in the Northern Rocky Mountain region.

Like thread 2, there were no statistical differences across time for debates surrounding the delisting method (thread 4). However, the tone of these debates did shift between time-periods. Before 2011, the debate surrounded the best way to conduct the delisting of gray wolves. In 2011, news covered the unconventional process, a rider on a Congressional budget bill. After 2011, political actors debated the legality of using the rider.

Finally, discourse surrounding management responsibility for continued gray wolf management varied across time (thread 5). After delisting, the majority of discourse focuses on who should be responsible for wolf management: states or the federal government, and whether the gray wolf can

be delisted without facing future threats ($EMP = 0.462, p < 0.001$).

Discussion

In summary, our analysis found that discourse varied substantively across political actors, with different stakeholders focused on different aspects of delisting, not just the science behind recovery. The executive branch focused on the content and viability of state management plans. The legislative branch also discussed state management plans, as well as the technical components of gray wolf protections under the ESA such as how distinct populations are defined and whether they have reached sufficient levels to warrant delisting. Meanwhile, the judicial actors concentrated primarily on the technical and procedural aspects of gray wolf protections under the ESA. Like the executive and legislative branches, members of the public and private sector actors discussed state management plans and ensuring wolf hunts and lethal removal of targeted animals as a means to respond to depredation. This group also questioned the motivations of decision-makers in the executive and legislative branches of government. Finally, like members of the public and private sector actors, NGO representatives scrutinized the political motivations of decision-makers, arguing that ESA decisions must be made on scientific bases, not politics. This group was also vocal in questioning which actor (federal or state managers) would be best suited to manage gray wolves.

Our study also found that the language used surrounding wolf delisting changed over time before, during, and after the 2011 delisting by Congress, showing how the debates over species management fluctuates. Before delisting, the discourse was focused on the suitability of state management plans. During delisting in 2011, the discourse turned to covering the ESA delisting process, and the unconventional delisting method applied. After the 2011 delisting, actors continue to debate the delisting method and present conflicting perspectives on population control. Proponents of relisting contend that current hunting seasons and state-sanctioned lethal control measures result in unsustainable mortality rates. Conversely, stakeholders in states such as Idaho argue that the gray wolf population has exceeded recovery goals, necessitating increased hunting quotas and more aggressive culling by state wildlife officials. This polarized discourse, in which conflicting data is presented, suggests that political considerations often overshadow scientific consensus in the formulation of gray wolf management policies. Our analysis also showed NGOs dominating the discourse in the media, rather than representatives from the USFWS and state wildlife agencies. This finding echoes a study of the political discourse around grizzly bear delisting, which found groups other than wildlife managers in the executive branch are increasingly the most vocal in the media realm (Mollett et al. 2025). Several phenomena might explain the hesitancy of agency scientists to speak vocally about a contentious topic such as wolf delisting. First is a concern that

Table 3. Sets of co-occurring words and narrative descriptions across time (before, during, and after 2011 delisting). 'This column represents the time periods when actors were talking about that topic significantly more than during other time periods.'

Thread of temporal political discourse	Semantic coherence (exclusivity)	Narrative description	Most common period(s) ¹	Example quote
Thread 1: state, plan, manage, federal, wildlife, delist, fish	-7.529012 (8.435724)	State management – content of plans: USFWS and the three states significantly divided regarding the content of proposed wolf management plans.	Before	'Federal officials have not moved forward with delisting because ... Idaho and Montana have submitted and received approval of their wolf management plans, but Wyoming's submitted plan was promptly rejected by the USFWS primarily because the plan allowed for shooting wolves on sight.' Before (Sewell 2005)
Thread 2: state, kill, manage, population, hunt, livestock, wildlife	-21.541585 (7.996671)	State management – hunting and lethal control: across all time periods, debate surrounds the interaction between wolves and livestock, and how states will manage wolf populations through hunting and lethal control by livestock producers.	Before, during, after	'With the rising population, state officials, including Idaho Gov Jim Risch and Montana Gov Brian Schweitzer, have been pushing the [USFWS] to remove [ESA] protections that officials say hamper control efforts aimed at stopping the predators that have spread into populated areas and are eating livestock as well as elk that are prized by hunters.' Before (Press 2006). 'Wyoming's proposal to classify wolves as trophy game subject to regulated hunting in northwest Wyoming ... hung up delisting in the state while Montana and Idaho inched toward taking over management of their populations.' During (SHOOT ON SIGHT 2011). 'Big Game Forever contends that out-of-control wolf populations are decimating big game herds in Idaho... That assessment is not generally supported by science' said Michael Robinson, the Center for Biological Diversity's expert on the gray wolf. 'That's a claptrap. Idaho has larger numbers of elk now than before the [1995 wolf] reintroduction.' After (Maffly 2020)
Thread 3: state, manage, species, endangered, delist, rule, federal	-18.978145 (8.581599)	ESA delisting success: vocal political actors frequently described delisting as a sign of the ESA's success.	During	'Interior Secretary Ken Salazar said ... wolves in the Northern Rockies are biologically recovered. Like other iconic species, such as the whooping crane, the brown pelican, and the bald eagle, the recovery of the gray wolf is another success story of the Endangered Species Act.' During (Puckett 2011)
Thread 4: delist, species, endangered, federal, bill, protect, wildlife	-26.695905 (7.554388)	The process of delisting and debate surrounding the method used: Before 2011, this thread discussed delisting wolves from the ESA. In 2011, news coverage focused on Congress's unprecedented delisting action. After 2011, media coverage focused on the legality of that decision.	Before, during, after	'This is good news for wolves, elk, rural communities, and hunters ... The Endangered Species Act was not meant to keep animals listed forever' Before . Idaho Fish and Game Director Cal Groen in response to Interior Secretary Ken Salazar affirming delisting decision (Wolf Delisting Rule Announced 2009). 'Sen Jon Tester (D-Mont) and Rep Mike Simpson (R-Idaho) today placed a rider on the must-pass federal budget bill that removes wolves... from the federal endangered species list and sets the stage for near-term delisting in Wyoming.' During (Tester, Simpson Sneak Wolf-Killing Rider Into Budget Bill 2011). 'A federal appeals court ruled Wednesday that Congress acted legally when it eliminated [ESA] protections for the Northern Rocky Mountain population of gray wolves and opened the door to wolf hunts.' After (Boxall 2012)
Thread 5: state, species, protect, wildlife, service, endangered, delist	-9.080791 (7.858680)	Management responsibility and state protections: delisting of the Rocky Mountain subpopulation, the debate remains prominent regarding state actions and whether the gray wolf should stay delisted.	After	'Days after Idaho Gov Brad Little signed a bill allowing the state to kill up to 90 percent of the state's wolves, more than 100 scientists, several from the Northwest, this week called upon Interior Secretary Deb Haaland and the US Fish and Wildlife Service to reinstate federal protections for gray wolves under the Endangered Species Act.' After (CBB 2024)

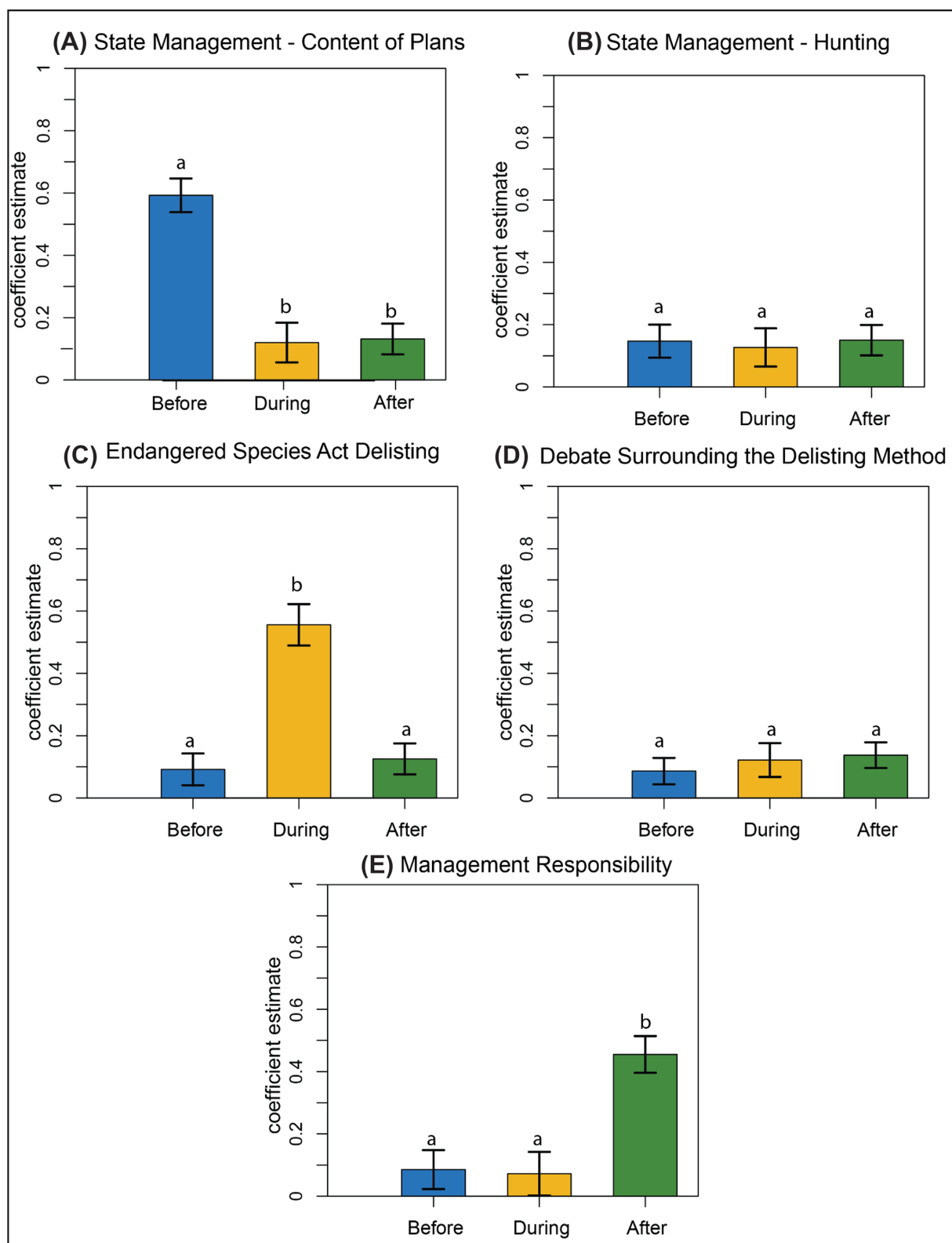


Figure 4. Predicted frequency of use of the five most common threads across time derived from the means parameterization regression. The bars represent the predicted mean frequency of the use of each topic (i.e. the coefficient estimate from the regression) for each time period. Confidence intervals around each bar indicate the 95% confidence interval around these mean frequencies. Letters above the bars denote statistically significant differences across time. Each panel (A–E) denotes the frequency of the statistical topic where the model identified co-occurring words for the dataset as a whole, and panels are labeled by the qualitative interpretation of these themes.

speaking out may be perceived as advocacy and be inappropriate for objective scientists (Crooks et al. 2022). Another potential component is that despite the best intentions of the ESA to make decisions based on empirical science, divisive species management topics, such as wolf delisting, are inherently political (Carolan 2008, Wyman 2012). While there is general scientific consensus that gray wolves should be relisted (Crooks et al. 2022), the controversies surrounding delisting are centered on political divides and state versus federal management disagreements (Martin 2020, Crooks et al. 2022). Therefore, while state or federal agency scientists may hesitate to make divisive statements, advocacy groups and NGOs with clear pro or anti-wolf missions are more empowered to make definitive statements in the media realm. These groups are frequently also engaged in lawsuits regarding delisting or relisting decisions, and legal advocates involved in the legal battles surrounding wolf delisting are likely to be highlighted in the media. For example, one article we reviewed focused on lawsuits by twelve different environmental groups (Knickerbocker 2008).

Another important finding of our analysis was that the way stakeholder groups talk about wolves differs substantially. Some political actors focused on the correct delisting process under the ESA statute. In contrast, others were concerned about their livelihoods, while certain groups engaged in political discourse because of their underlying values towards wolves as a species. The tension caused by underlying values in wolf management conflicts is described by USFWS wolf biologist Ed Bangs as surrounding symbolism and feelings towards wolves rather than the animals themselves (McQuillan 2006). Previous researchers have reaffirmed this sentiment, asserting that wolf delisting and public opinions surrounding it are inherently value-based (Nie 2002). These findings suggest that part of the contention surrounding gray wolves and other high-profile listed species may be due to various stakeholders having different, and not always overlapping, conversations. Adaptive governance literature highlights the importance of incorporating various perspectives beyond the best available science in decision-making (Brunner 2005, Chaffin et al. 2014). Further, increased dialogue between stakeholders in contentious carnivore management has been found to improve trust and cooperation (Salvatori et al. 2021). The threads of political discourse identified in this study could help to develop messaging that speaks to various stakeholder groups' concerns and guide collaborative problem-solving processes.

Dissatisfaction from state governments over the extent of control the ESA provides the federal government, even within state boundaries, is not new. Previous research has illustrated how low state engagement with the ESA has led to low popularity of the ESA in certain states (Nagle 2017), and the 2011 gray wolf delisting has been used to illustrate this frustration (Perry 2012). Our analysis of the threads of political discourse used by political actors supports this assessment. For example, state management of wolves was a key theme across both structural topic model groups (i.e. political actors and time). While there was debate within stakeholder group

categories, many state government representatives and members of certain public and private sector groups argued for wolf delisting. Vocal members of these stakeholder groups advocated for the ability of state wildlife agencies and state residents to mitigate livestock and ungulate conflicts using hunting and lethal control measures. Perry (2012) argues that the solution to this frustration is more engagement between states and the federal government through Section 6 of the ESA: 'Delegating greater ESA authority to the states under Section 6 would reduce political opposition against the Act by changing its image as a federal intrusion on traditional state authority and allowing state agencies with knowledge of local conditions to make conservation decisions that are less burdensome to citizens' (p. 442).

Another notable finding of this study was how the 2011 delisting decision has continued to impact political discourse surrounding gray wolf delisting. For example, following the 2011 delisting, there has been more focus on the decision's legality and concern about the legislative branch intervening in ESA management. A previous study analyzing public discourse surrounding wolf delisting in Idaho between 1995 and 2009 found delisting events were turning points in how the public discussed wolf issues (Killion et al. 2019). Additionally, because Congress, rather than wildlife management agencies, implemented the 2011 delisting decision, this action could perpetuate doubts regarding endangered species recovery science. Previous studies assessing rhetoric surrounding grizzly bear delisting found that scientific uncertainty, or questioning recovery science, influenced public discourse (Parker and Feldpausch-Parker 2013, Mollett et al. 2025). The findings of our study suggest that high-profile carnivore delisting decisions impact endangered species management across the US, by influencing rhetoric and obscuring the findings of recovery scientists.

We also observed how states designed their management plans and implemented delisting in the Northern Rocky Mountain states continue to influence nationwide gray wolf management debates. A new wave of political discourse surrounding wolf delisting has arisen in the past five years following a 2020 nationwide delisting (reversed in 2022) (Ferri 2022). Our article review showed that opponents of these recent delisting proposals use the Northern Rocky Mountain delisting as a platform and justification for opposition. For example, John Vucetich, a Professor at Michigan Technological University, expressed uncertainty regarding nationwide delisting proposals, using the 'hostile treatment of wolves' in Wyoming, Montana, and Idaho as a reason to retain wolves on the endangered species list (CBB 2024).

Like most research, this study has several methodological limitations. For example, using media as our scope allows us to observe patterns and discourse across groups and time. However, because the information used is secondhand, there are potential concerns of bias from the reporters and news outlets covering the topic. This limitation is minimized by using ProQuest and a systematic article selection method to ensure news sources and press releases from across the spectrum of wolf attitudes were included. Additionally, in

this study we did not assess spatial or temporal outlet concentration or variation between types of sources. Future research using these methods could assess how discourse varies between types of news sources (e.g. clustering outlets and assessing news vs press releases) or spatial outlet concentration (e.g. differences between regional or national news sources). Finally, a previous study using this research method to assess the political discourse surrounding grizzly bear delisting identified that this type of media content analysis has a potential political platform effect limitation (Mollett et al. 2025). In other words, certain actors, such as the legislative branch and NGOs, might use the media as a platform and therefore be more vocal than other voices. We acknowledge this potential limitation but argue that understanding how political actors politicize the ESA is a key goal of this research that would not be apparent by solely reviewing management documents or interviewing scientists and managers involved in wolf delisting.

Conclusion

Nationwide, debates and fluctuations in gray wolf protections continue. The proliferation of court battles and frequent delisting and relisting decisions highlights the role that politics continue to play in wolf management. In this study we found analyzing media sources to be an effective tool for synthesizing divergent stakeholder perspectives. Managers and decision-makers can incorporate these threads of political discourse into collaborative processes to ensure the values underlying endangered species perspectives of stakeholders are broadly represented. Additionally, in this study, we illuminate how the shift in ESA delisting authority from agency wildlife managers (i.e. scientists hired in the executive branch) to Congress (i.e. elected politicians) has had long-lasting impacts on both ESA implementation and stakeholder perceptions of delisting decisions. While our study focuses on the discourse surrounding gray wolf delisting, the findings of this study can be expanded beyond wolves to threatened and endangered species management generally.

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Author contributions

Iree Wheeler: Writing – Original draft (lead); Formal analysis (supporting); Writing – Review and editing (lead) **Paisley Blair:** Writing – Original draft (supporting); Writing – review and editing (equal). **Noah Creany:** Visualization (equal); Writing – review and editing (equal). **Hannah Manire:** Data curation (equal). **Daniel Morris:** Writing – review and editing (equal). **Sofia Mollett:** Data curation (equal). **Stelle S. Searce:** Data curation (equal). **Temple Stoellinger:** Conceptualization (equal); Methodology (equal); Writing – review and editing (equal). **Janna R.**

Willoughby: Conceptualization (equal); Formal analysis (lead); Methodology (equal); Writing – review and editing (equal). **Kelly Dunning:** Conceptualization (equal); Methodology (equal); Project administration (lead); Writing – review and editing (equal).

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Data availability statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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